॥ अप्पो दीपो भव ॥
महात्मा गांधी मिशन विश्वविद्यालय
MGMUNIVERSITY

# School of Engineering and Technology

# Where Technology Meets Tomorrow

UG 4 Year - B.Tech Programs after HSC

UG 6 Year - B.Tech Integrated Programs after SSC

Diploma Programs

You’re not just studying engineering. You’re engineering the future. With cutting-edge labs, future-ready programs, and a legacy of innovation, we empower you to design solutions that move the world forward.

Students at SOET have access to some of the most advanced facilities in the region. From building intelligent machines in the Industrial Robotics Lab to exploring sustainable mobility in the Electric Vehicle Lab, every space is designed to spark innovation and hands-on learning. Whether you're prototyping drones, designing Industry 4.0 systems, or launching your startup at the Innovation Incubation and Research Center, you'll find the tools and mentorship to go beyond the classroom.

Plus, a vibrant campus life with hostels, sports, and fitness facilities ensures holistic growth beyond academics.

- NEP 2020 Aligned Curriculum
- Outcome Based Education System
- Qualified and Experienced Faculty
- Innovative Teaching Methodologies like Problem Based Learning, Project Based Learning, Experiential Learnin
- Entrepreneurship and Startup Cell
- Well Equipped Laboratories
- Mechatronics and Industry 4.0 Lab
- Industrial Robotics Lab
- Electrical Vehicle Lab

# About SOET

The School of Engineering and Technology (SOET) at MGM University is a pioneer in offering Integrated six year B.Tech. Programs (after SSC) and Four-Year B.Tech. Programs (after HSC) in the Marathwada region. The Integrated program allows students to begin their engineering education right after Class 10, offering an early advantage and extended time for hands-on learning, innovation, and deep exploration of engineering fundamentals. This structure empowers young aspirants to build a strong foundation and stay ahead in the competitive landscape. SOET's integrated curriculum is designed to nurture curiosity, develop critical thinking, and prepare students for advanced industry demands from an early stage.

At the same time SOET offers focused Four-Year B.Tech. programs after Class 12, catering to students who aspire to pursue professional engineering education immediately after HSC. These programs are aligned with the latest industry trends and include emerging domains such as Computer Science and Engineering, Artificial Intelligence & Robotics, Cyber security, Semiconductor along with core experience program in Electrical, Robotics & Automation, Internet of Things (IoT), and Blockchain Technology. The teaching and learning process is Project-Based backed by Research, and is powered by Collaborative methods.

Students benefit from internship and placement opportunities through active collaborations with industries. With a legacy of over 42 years of MGM Trust, SOET is committed to producing future-ready engineers through holistic education, modern infrastructure and a strong emphasis on innovation and ethics.

# About MGM University

MGM University, established by the revered Mahatma Gandhi Mission Trust, a pioneer with four decades of remarkable legacy in education, research and service, is a State Private University, approved by Government of Maharashtra MGM University Act No. XXVI of 2019. It has the 2(f) status granted by the University Grants Commission (UGC) of India.

# UG 4 Year B.Tech Programs after HSC

## Computer Science and Engineering (IoT, Cyber Security Including Blockchain Technology)

**Core Courses**

- Embedded systems and IoT device programming
- Cybersecurity fundamentals: cryptography, network security, ethical hacking
- Blockchain principles: decentralized systems, smart contracts
- Hands-on with IoT platforms like Raspberry Pi
- Real-world applications in smart cities, healthcare, and industry 4.0

**Career Opportunities**

- IoT Developer/Solution Architect
- Cyber Security Analyst/Ethical Hacker
- Blockchain Developer/Consultant
- Security Auditor/Network Administrator
- Opportunities in smart infrastructure, defense, fintech, and healthcare
- S/W Developer

## Computer Science and Engineering (Cyber Security)

**Core Courses**

- Network security, cryptography, and risk management
- Cyber laws, digital forensics, and ethical hacking
- Hands-on labs for malware analysis and penetration testing
- Security compliance frameworks (ISO, NIST)
- Collaboration with government and private cyber defense initiatives

**Career Opportunities**

- Cybersecurity Analyst/Engineer
- Penetration Tester/Ethical Hacker
- Digital Forensics Expert
- Security Operations Center (SOC) Analyst
- High demand in banking, e-commerce, IT services, and government sectors
- S/W Developer

## Electronics Engineering (VLSI Design & Technology)

**Core Courses**

- VLSI design using HDL (Verilog/VHDL)
- Semiconductor device physics and fabrication processes
- Design tools like Cadence, Synopsys, Xilinx
- Focus on low power, high-speed chip design
- Cleanroom labs and embedded systems integration

**Career Opportunities**

- VLSI design Engineer
- ASIC/FPGA Developer
- Semiconductor Fabrication Engineer
- Embedded Systems Engineer
- Careers in chip design companies like Intel, Qualcomm, AMD, and TSMC

## Robotics and Automation

**Core Courses**

- Robotic kinematics, control systems, and path planning
- Automation technologies: PLC, SCADA, HMI
- AI integration with robotics (vision, learning, speech)
- Hands-on labs with robotic arms, drones, and mobile robots
- Applications in manufacturing, logistics, defense, and healthcare

**Career Opportunities**

- Robotics Engineer
- AI Research Scientist
- Autonomous Systems Developer (e.g., self-driving cars, drones)
- Automation Engineer
- Roles in healthcare, defense, manufacturing, and logistics

## Electrical Engineering

**Core Courses**

- Electrical machines, power systems, and Energy conversion
- Power electronics and high-voltage engineering
- Renewable energy systems (solar, wind)
- Control systems and smart grid technology
- Simulation tools like /Simulink and ETAP

**Career Opportunities**

- Electrical Engineer (Power Distribution, Generation)
- Control System Engineer
- Energy Consultant/Renewable Energy Expert
- Electrical Design Engineer
- Opportunities in government utilities,
- infrastructure, manufacturing, and railways

# UG 6 Year

# B.Tech Integrated Programs after SSC

## Computer Science and Engineering (IoT, Cyber Security Including Blockchain Technology)

**Core Courses**

- Embedded systems and IoT device programming
- Cybersecurity fundamentals: cryptography, network security, ethical hacking
- Blockchain principles: decentralized systems, smart contracts
- Hands-on with IoT platforms like Raspberry Pi
- Real-world applications in smart cities, healthcare, and industry 4.0

**Career Opportunities**

- IoT Developer/Architect
- Cybersecurity Analyst/Engineer
- Blockchain Developer/Consultant
- Network Security Specialist
- Embedded Systems Engineer
- Roles in smart cities, healthcare, automotive, and fintech industries

## Computer Science and Engineering (Data Science)

**Core Courses**

- Programming in Python, R, and SQL
- Machine learning, deep learning, and artificial intelligence
- Data visualization, data mining, and cloud computing
- Big data tools like Hadoop, Spark, and TensorFlow
- Industry-oriented projects and internships

**Career Opportunities**

- Data Scientist/Analyst
- Machine Learning Engineer
- Business Intelligence Developer
- AI/ML Researcher
- Data Engineer
- Opportunities in finance, e-commerce, health tech, and more

## Robotics and Artificial Intelligence

**Core Courses**

- Robotics design, mechatronics, and automation
- AI algorithms, computer vision, and natural language processing
- Control systems, embedded programming, and sensor integration
- Hands-on robotics labs and interdisciplinary projects
- Exposure to neural networks and reinforcement learning

**Career Opportunities**

- Robotics Engineer
- AI Research Scientist
- Autonomous Systems Developer (e.g., self-driving cars, drones)
- Automation Engineer
- Roles in healthcare, defense, manufacturing, and logistics

## Civil Engineering

**Core Courses**

- Core subjects: structural engineering, geotechnical engineering, transportation, environmental engineering
- Surveying, construction management, and fluid mechanics
- Use of software like AutoCAD, STAAD Pro
- Site visits, lab work, and internships with construction firms
- Sustainability and green building practices

**Career Opportunities**

- Civil Engineer (Site/Design)
- Structural Engineer
- Transportation Planner
- Environmental Engineer
- Project Manager
- Government jobs (PWD, CPWD, Railway, Municipal
- Corporations), private sector, or entrepreneurship

# Diploma Programs

## 2 Year

### Diploma in Fire Service and Engineering

**Core Courses**

- Study of fire science, fire behavior, and firefighting equipment
- Fire detection systems, sprinkler systems, and
- emergency evacuation planning
- Training in rescue operations, disaster management, and first aid
- Fire safety laws, building codes, and fire control room operations
- Practical fire drills and live training exercises

**Career Opportunities**

- Fire Officer/ Firefighter
- Fire & Safety Engineer
- Fire Inspector/ Fire Prevention Specialist
- Fire Protection System Designer
- Roles in fire departments, airports, industrial
- complexes, oil & gas facilities, and metro rail projects

## 1 Year

### Advanced Diploma in Industrial Safety

**Core Courses**

- Study of safety management, accident prevention, and hazard control
- Focus on legal frameworks like the Factories Act and Safety Standards (OSHA, ISO)
- Training in fire safety, emergency planning, and industrial hygiene
- Risk assessment, audits, and safety inspection methods
- Industrial visits and real-time case studies

**Career Opportunities**

- Safety Officer/ Safety Supervisor
- HSE (Health, Safety & Environment) Officer
- Safety Auditor/ Compliance Officer
- Safety Consultant in manufacturing, oil & gas, construction, and power sectors
- Opportunities in both public and private industries, with high demand abroad

# Student Activities

EXPERTS LIVE INDIA CONFERENCE 2025

PROJECT EXHIBITION

ZESTOPIA 2025 CULTURAL EVENT

TRAINING PROGRAM ON ROBOTICS AND AUTOMATION

# Eligibility

## B.Tech (Integrated)

- SSC Passed
- Min 50% for Open and Min 45% for Reserved Category Candidate
- MGMU-CET

## Lateral Entry

- Direct Second Year after HSC Passed

## B.Tech

- HSC Passed
- Min 45% for Open and Min 40% for Reserved Category
- Candidate should have appeared in JEE(Mains)/
- MHT-CET/ PERA-CET/ MGMU-CET (With non zero score)

## Lateral Entry

- Direct Second Year after Diploma Course in Engineering and Technology/ B.Sc. Degree/ D. Voc.

# Admission Procedure

01 Fill Application on admissions.mgmu.ac.in

02 Appear for MGMU CET

03 Check Merit List Displayed on Dashboard

04 Verification & Submission of Documents

05 Processing of Fees

06 Admission Successfully Processed

INDUSTRIAL VISITS

NATIONAL SERVICE SCHEME (NSS) ACTIVITIES

DISCUSSION WITH AUTHOR OF BOOK "GARAGE TO GLOBAL"

SPORTS ACTIVITIES

# Collaborations

# Campus Placement Prime Recruiters

MGMUNIVERSITY Mahatma Gandhi Mission University

**Address**
Mahatma Gandhi Mission University, MGM Campus,
N-6 CIDCO, Chhatrapati Sambhajinagar - 431003

**Website**
soet.mgmu.ac.in

**Phone**
+91 9371714253

**Email**
soet@mgmu.ac.in

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